

MIDWEST GENETICS LABORATORY

Clinical Molecular Genetics

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March 10, 2026

VARIANT RECLASSIFICATION NOTIFICATION

Patient: Marcus D. Whitfield **MRN:** PED-2025-04471
DOB: 03/03/2016 **Original Report Date:** 06/01/2025
Ordering Provider: Anita Sharma, MD — Cook County Pediatrics

Dear Dr. Sharma,

We are writing to inform you that a variant previously identified in your patient Marcus D. Whitfield has been reclassified based on new evidence.

VARIANT DETAILS

Gene: NF1 **Variant:** c.2033A>G (p.Asn678Ser)

Previous Classification: Variant of Uncertain Significance (VUS) — reported 06/01/2025

Updated Classification: Likely Pathogenic — effective 03/10/2026

BASIS FOR RECLASSIFICATION

Since the original report, additional evidence has become available supporting the pathogenicity of this variant. Specifically, the variant has now been reported in two additional unrelated individuals with clinical NF1, functional studies have demonstrated reduced GTPase-activating protein (GAP) activity of the neurofibromin protein harboring this substitution, and the variant meets PM1, PM2, PP3, and PS3 criteria per updated ACMG/AMP classification framework.

CLINICAL IMPLICATIONS

This reclassification confirms a molecular diagnosis of Neurofibromatosis Type 1 in this patient. The current managing clinician should ensure the following per ACMG/AMP guidelines for confirmed NF1:

- Annual ophthalmologic examination including slit-lamp evaluation for Lisch nodules and assessment for optic pathway glioma.
- Annual blood pressure monitoring for renal artery stenosis.
- Monitoring for scoliosis, tibial dysplasia, and other skeletal manifestations.
- Brain MRI as clinically indicated. If prior MRI did not include optic pathway sequences, repeat imaging with dedicated optic pathway protocol is recommended.
- Genetic counseling for the patient and family regarding inheritance pattern, recurrence risk, and long-term surveillance recommendations.

This letter is being sent to the ordering provider of record (Dr. Sharma). If the patient has transferred care, please forward this notification to the current managing clinician.

Patricia Holden, MD, PhD, FACMG

Laboratory Director
Midwest Genetics Laboratory